MILITARY SPECIFICATION

SHIPPING AND STORAGE CONTAINER, CARTRIDGE, 20 MILLIMETER: XM528E1

1. SCOPE

1.1 This specification covers the XM528E1 container used for packaging linked 20mm ammunition for the Vehicle Rapid Fire Weapon System - Interim (Cannon, Automatic, 20mm, XM139).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-C-490	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coating

STANDARDS

Federal

Federal Test Method Std. No. 151 - Metals; Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-406	Visual Inspection Standards for Terne Plate Cans and Steel Boxes Used in Small Arms Ammunition Packaging
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Per cent Defective

DRAWINGS

Munitions Command

FSC 8140

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F10542308	Shipping and Storage Container, Cartridge, 20 Millimeter: XM528E1
F10542342	Box, Shipping: Shipping and Storage Container, Cartridge, 20 Millimeter: XM528E1
IEL-10542308	Index of Inspection Equipment Lists for: Shipping and Storage Container, Cartridge, 20 Millimeter: XM528E1

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General - Materials, components, and sub-assemblies shall comply with the applicable drawings, referenced specifications, and the following:

3.2 Material - The contractor shall be responsible for selecting from the classification list, contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.3 First article sample - Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Manufacturing process - Components and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without his prior approval.

3.5 Assembly - Components and assemblies shall be combined to produce an assembly which conforms to the requirements of Drawing F10542308.

3.6 Protective finish - When protective finish is required by component or assembly details, finish requirements shall apply to all surfaces of the assembled container regardless of contour configuration.

3.6.1 Adhesion - The protective finish shall not peel, flake, or fail to adhere after 24 hours of drying.

3.6.2 Salt Spray - The finished container shall withstand 120 hours exposure to a salt spray of a five percent sodium chloride solution.

3.7 Functioning

3.7.1 Cover assembly - The cover of the container shall open, close, and be removed without binding or requiring exertion of undue force. The gasket shall not stick to top edges of the container, be cut or split, nor shift within the gasket retainer when the cover is removed.

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3.7.2 Hasp and latch - Mating parts of the body hasp and the latch shall meet without requiring deformation of any container part, and the latch shall close and open freely. When closed, the latch of the assembled container shall remain closed until manually opened.

3.7.3 Elevated temperature storage - Following storage of the container in its closed position at an air temperature of 165 plus or minus 2 degrees Fahrenheit (F) for a period of 24 hours and subsequent return to ambient temperature, the cover shall function as required in 3.7.1.

3.8 Airtightness - The assembled container shall not release more than 4 bubbles of air when subjected to an air pressure differential of 3 pounds per square inch (PSI) for 15 seconds.

3.9 Welds and hardware components

3.9.1 Cover handle assembly - The cover handle of the container shall withstand a pull of 300 pounds for a period of one minute without separation from the cover or failure of either link, clip, or any weld.

3.9.2 Body and handle assemblies - Each body and handle assembly of the container shall withstand a pull of 100 pounds for a period of one minute without separation from the hinge or hasp or failure of the handle or grip.

3.9.3 Latch, cover, and hinge assembly - The latch, cover, and hinge assembly shall withstand a pull of 500 pounds for a period of one minute without breakage or distortion of any of the components or welds.

3.9.4 Hasp assembly - The assembly of the hasp to the container shall withstand a pull of 1,000 pounds for one minute without failure of the hasp or weld.

3.10 Interchangability (cover and body assembly) - Fabrication of body assemblies and cover assemblies shall be such that these assemblies can be interchanged at random to produce containers which comply with the requirements of 3.7.1, 3.7.2, and 3.8.

3.11 Workmanship - The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

3.11.1 Metal defects - The containers shall be free of burrs, rough spots, sharp projections, cracks, splits, buckles, or sintered surfaces.

3.11.2 Foreign matter - The containers shall be free of corrosion, chips, dirt, grease, and other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the

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contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions - Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample

4.2.1. Initial production sample - At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of:

a. Ninety-six (96) assembled painted containers.
b. Five (5) sets of unassembled and unpainted parts.

The sample shall be manufactured using the same materials, equipment, processes, and procedures as used in regular production.

4.2.1.1 Examination and test - After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications, at a government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure - Failure of the sample to meet the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions

4.3.1 Lot

4.3.1.1 Submittal of product - The contractor shall submit the product in accordance with Standard MIL-STD-105.

4.3.1.2 Size of lot - The lot size shall be as agreed upon between the contractor and the contracting officer provided the formation of the lot is in accordance with Standard MIL-STD-105.

4.3.2 Examination - Examination for major and minor defects shall be performed on a class basis, using the applicable sampling plans and acceptance criteria of Standard MIL-STD-105 for the Acceptable Quality Levels (AQL's) prescribed in the classification of defects. All non-conforming containers and packed shipping boxes shall be rejected.

4.

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4.3.2.1 Classification of defects - The classification of defects and applicable AQL's shall be as follows:

4.3.2.1.1 Shipping and Storage Container, Cartridge, 20 millimeter: XM528E1 (see Drawing F10542308) 1/

Category, AQL and defects	Method of Inspection
Critical: None defined	
Major (AQL 1.00%)	
101. Incomplete manufacture	Visual
102. Incorrect assembly	Visual
103. Burrs, rough spots, sharp projections, cracks, splits, buckles	Visual
104. Incomplete or inadequate paint coverage	Visual
105. Length, outside, max	Gage
106. Width, outside, max	Gage
107. Height, outside, max	Gage
108. Length, body assembly inside, min	Gage
109. Width, body assembly inside, min	Gage
110. Height, body assembly inside, min	Gage
111. Concavity or convexity of cover assembly	Gage
112. Concavity or convexity of body assembly	Gage
Minor (AQL 2.50%)	
201. Chips, dirt, grease, other foreign matter, and other defects	Visual
202. Surface imperfections, excessive paint, paint not dry	Visual
203. Manufacturer's identification missing, or incomplete	Visual

4.3.2.1.2 Box, Shipping: Shipping and Storage Container, Cartridge, 20 millimeter: XM528E1 (see Drawing F10542342)

Category, AQL and defects	Method of Inspection
Critical: None defined	
Major: None defined	
Minor (AQL 2.50%)	
201. Contents loose	Manual/visual

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202.	Contents incorrect	Visual
203.	Separator missing	Visual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Box inadequately or improperly sealed	Visual
207.	Markings missing, incorrect or illegible	Visual

1/ In the examination for visual defects MIL-STD-406 shall be used as a basis of comparison for acceptance, where applicable, except that all defects classified as "Incidental" in MIL-STD-406 shall be considered permissible.

4.3.3 Tests - The following tests shall be performed on each lot in accordance with the provisions of 4.4.

4.3.3.1 Protective finish - Failure on one or more of these tests shall be without penalty to the lot, but the manufacturer shall be required immediately to correct his surface preparation and/or painting practice, as pertinent, to comply with applicable specification.

4.3.3.1.1 Adhesion - A sample of 3 containers shall be subjected to the tape test. Test failure is defined as one or more test containers showing any bare metal or underlying phosphate coating on either the cover or the body assembly.

4.3.3.1.2 Salt spray - A sample of 3 containers shall be subjected to the salt spray test. Test failure is defined as one or more test containers showing rust spots, wrinkles, or blisters on any of the body or cover assembly surfaces, or evidence of rust creepage extending more than 1/8 inch on either side of the scribe marks, or rusting of part of the hardware subject to pressure or abrasion in normal use, or rust exceeding the permissable standards of MIL-STD-406, or enamel softer than unexposed enamel after air drying for 24 hours following the test.

4.3.3.1.3 Film thickness - A sample of 3 containers shall be checked for film thickness. Test failure is defined as one or more test containers not conforming to the drawing requirement.

4.3.3.2 Functioning

4.3.3.2.1 Cover assembly - Each container of the examination sample (4.3.2.1.1) shall be tested for compliance with the requirement of 3.7.1. Failure of the sample to meet acceptance criteria of MIL-STD-105 for an AQL of 1.0% shall be cause for rejection of the lot.

4.3.3.2.2 Hasp and latch - Each container of the examination sample (4.3.2.1.1) shall be tested for compliance with the requirement of 3.7.2. Failure of the sample to meet the acceptance criteria of MIL-STD-105 for an

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AQL of 1.0% shall be cause for rejection of the lot.

4.3.3.2.3 Elevated temperature storage - A sample of 35 acceptable containers shall be selected from the examination sample (4.3.2.1.1) and tested for compliance with 3.7.3. Failure of any container to comply with the requirement shall be cause for rejection of the lot.

4.3.3.3 Airtightness - A sample of 35 acceptable containers shall be selected from the examination sample (4.3.2.1.1) and tested for compliance with 3.8. Failure of any container to comply with the requirement shall be cause for rejection of the lot.

4.3.3.4 Welds and hardware components - The same 35 containers used for the airtightness test (4.3.3.3) shall be tested for compliance with the requirements of 3.9. Failure of any component to meet the applicable requirement shall be cause for rejection of the lot.

4.3.3.4.1 Disposition of tested boxes - Those containers which have successfully passed the tests prescribed in 4.3.3.4 may be returned to the lot provided they are subsequently tested for airtightness (4.3.3.3) and ambient temperature functioning (4.3.3.2.1 and 4.3.3.2.2) and are found acceptable.

4.3.3.5 Gasket compression test - A sample of 10 acceptable containers shall be selected from the examination sample (4.3.2.1.1) and tested for Gasket Compression. The variables sampling plan of MIL-STD-414 example B-3 for an AQL of 1.00 percent shall be used. Failure of any container to comply with the drawing requirements or failure of the sample to meet the criteria of the variables sampling plan, shall be cause for rejection of the lot.

4.3.3.6 Interchangability (cover and body assembly) - A sample of 5 acceptable containers shall be selected from the examination sample (4.3.2.1.1) and tested for compliance with the requirements of 3.10. Failure of more than one assembled container to comply with the requirements of 3.7.1 and 3.7.2 or failure of more than one assembled container to comply with the requirement of 3.8 shall be cause for rejection of the lot.

4.3.4 Inspection equipment - The examination and tests shall be made using equipment listed on IEL-10542308.

4.4 Test methods and procedures

4.4.1 Protective finish

4.4.1.1 Adhesion - This test shall be conducted in accordance with the paragraph of TT-C-490 applicable to the test for paint adhesion. Three places, randomly located, on the surfaces of each sample container body and cover

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assembly shall be tested to assure adequate paint adhesion on all surfaces.

4.4.1.2 Salt spray - The sample shall be tested as specified in Standard Federal Test Method Std No. 151, Method 811, for a period of 120 hours with a five percent solution of sodium chloride. Samples shall be so positioned that the hinged edge of each assembled container is elevated approximately 15 degrees from the horizontal and rests upon a wooden support while the cover assembly face is parallel to the principal direction of horizontal flow of the fog through the chamber. Prior to exposure, cross marks (x) approximately three inches long and penetrating to base metal shall be made with a machinists scriber (or equivalent) across the flat exterior portion of each side, the bottom, the hinge end, and the cover.

4.4.1.3 Film thickness - Measurements shall be made at a random location on the exterior of each body assembly surface, a cover assembly surface and an accessible hardware surface. The test shall be conducted in accordance with paragraphs of TT-C-490 applicable to the test for paint thickness.

4.4.2 Functioning - The sample containers, while at ambient temperature, shall be manually and visually examined to assure compliance with the requirements of 3.7.1 and 3.7.2.

4.4.3 Elevated temperature storage - The sample containers shall be conditioned as prescribed in 3.7.3 and tested in accordance with 4.4.2.

4.4.4 Airtightness - The sample containers shall be tested by submerging in ambient temperature water, maintaining them for 15 seconds at an air pressure differential of 3 pounds per square inch in excess of the outside pressure using an approved vacuum testing machine.

4.4.5 Welds and hardware components

4.4.5.1 Cover handle assembly - The cover handles shall be tested by clamping the assembled, latched containers in prescribed fixture and slowly applying the force specified in 3.9.1 and maintaining such force for one minute.

4.4.5.2 Body end handle assemblies - The end handles shall be tested by clamping the body assembly in the prescribed fixture and slowly applying the force specified in 3.9.2 to each end handle and maintaining such force for one minute.

4.4.5.3 Latch, cover and hinge assembly - The assembly shall be tested in the prescribed fixture with the container resting on its bottom and with the cover raised at a right angle. The container shall be clamped securely with the body assembly supported to prevent distortion or collapse by means of a snug fitting wood filler block. The vertical force prescribed in 3.9.3 shall be applied slowly and maintained for a period of one

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minute with bearing against the face of the latch, when the latch is positioned parallel to the bottom of the container.

4.4.5.4 Hasp assembly - The container shall be clamped in the prescribed fixture, resting on its bottom and with cover assembly removed. The force specified in 3.9.4 shall be slowly applied in a vertical direction, with bearing against the underside of the offset of the hasp over the entire surface available, and maintained for a period of one minute. The container body shall be supported, to prevent distortion or collapse, by means of a snug fitting wood filler block.

4.4.6 Gasket compression - Gasket compression shall be measured using gages specified on the applicable inspection equipment list and in accordance with the instructions provided thereon. For each sample container, the average of the four compression measurements shall be computed. The average values thus obtained shall be used in the application of the procedures of MIL-STD-414.

4.4.7 Interchangability - The cover assembly shall be removed from each of the completely assembled containers and be reassembled in turn on each of the different container bodies from which the cover assembly was removed. Each reassembled container shall be tested for Functioning (4.4.2) and Airtightness (4.4.4).

5. PREPARATION FOR DELIVERY

5.1 Packing - Six assembled containers with latches closed shall be packed in a fiberboard box as shown on Drawing F10542342.

5.2 Marking - Fiberboard boxes shall be marked in accordance with Drawing F10542342.

6. NOTES

6.1 Ordering data - Invitation for bids and contracts or orders will specify the following.

6.1.1 Title, number, and date of this specification.

6.1.2 Place of inspection, if not at place of manufacture.

6.1.3 First article sample requirements. (See 4.2)

6.1.4 Provision for the submission of acceptance inspection reports containing final inspection results for each lot of containers presented to the Government.

6.1.5 Requirement for contractor to provide and maintain an inspection

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system in accordance with MIL-I-45208, Inspection System Requirements.

6.2 Intended use - The XM528E1 container is intended for packaging linked 20mm ammunition for the Vehicle Rapid Fire Weapon System - Interim (Cannon, Automatic, 20mm, XM139).

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No:

8140-A039

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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